9 A radon-222 ($^{222}_{86}Rn$) nucleus, originally at rest, spontaneously decays to form a polonium-218 ($^{218}_{84}Po$) nucleus and an alpha particle. It may be assumed that no gamma ray is emitted.

(a) Explain what is meant by *spontaneous*.

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..... [1]

The rest masses of the nuclei are shown in Fig. 9.1.

$^{222}_{86}Rn$	222.0176 u
$^{218}_{84}Po$	218.0090 u
alpha particle	4.0026 u
proton	1.00727 u
neutron	1.00866 u

Fig. 9.1

(b) **(i)** Calculate the total kinetic energy of the decay products.

total kinetic energy = ...J [3]

(ii) Describe the subsequent motion of the decay products. Explain your answer with reference to the principle of conservation of momentum.

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..... [2]

(iii) Show that the ratio $\dfrac{\text{kinetic energy of alpha particle}}{\text{kinetic energy of Po-218 nucleus}} \approx 54.5$.

[1]

(c) **(i)** Calculate the value of mass defect per nucleon $\left(\text{i.e. } \dfrac{\text{mass defect}}{\text{number of nucleons}}\right)$ for Radon-222. Leave your answer in terms of atomic mass units (u).

mass defect per nucleon for Radon-222 = ... u [3]

(ii) The mass defect per nucleon for Polonium-218 has a value of 8.08312×10^{-3} u. With reference to your answer in **(c)(i)**, explain whether Polonium-218 or Radon-222 is more stable.

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..... [3]

(d) Radon-222 has a half-life of 3.8 days.

(i) State what is meant by *half-life*.

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..... [1]

(ii) Calculate the probability of a given radon-222 nucleus decaying per second.

probability = ... s^{-1} [2]

(iii) A student stated that "radioactive materials with a short half-life always have a high activity". Discuss whether the student's statement is valid.

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..... [1]

(e) A sample of Radon-222 was carefully measured out and sealed in a container. The rate of radioactive decay was measured using an accurate instrument, taking into account background radiation. The number of alpha particles detected was significantly higher than expected. State what this suggests about the stability of Polonium-218. Explain your answer.

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..... [3]